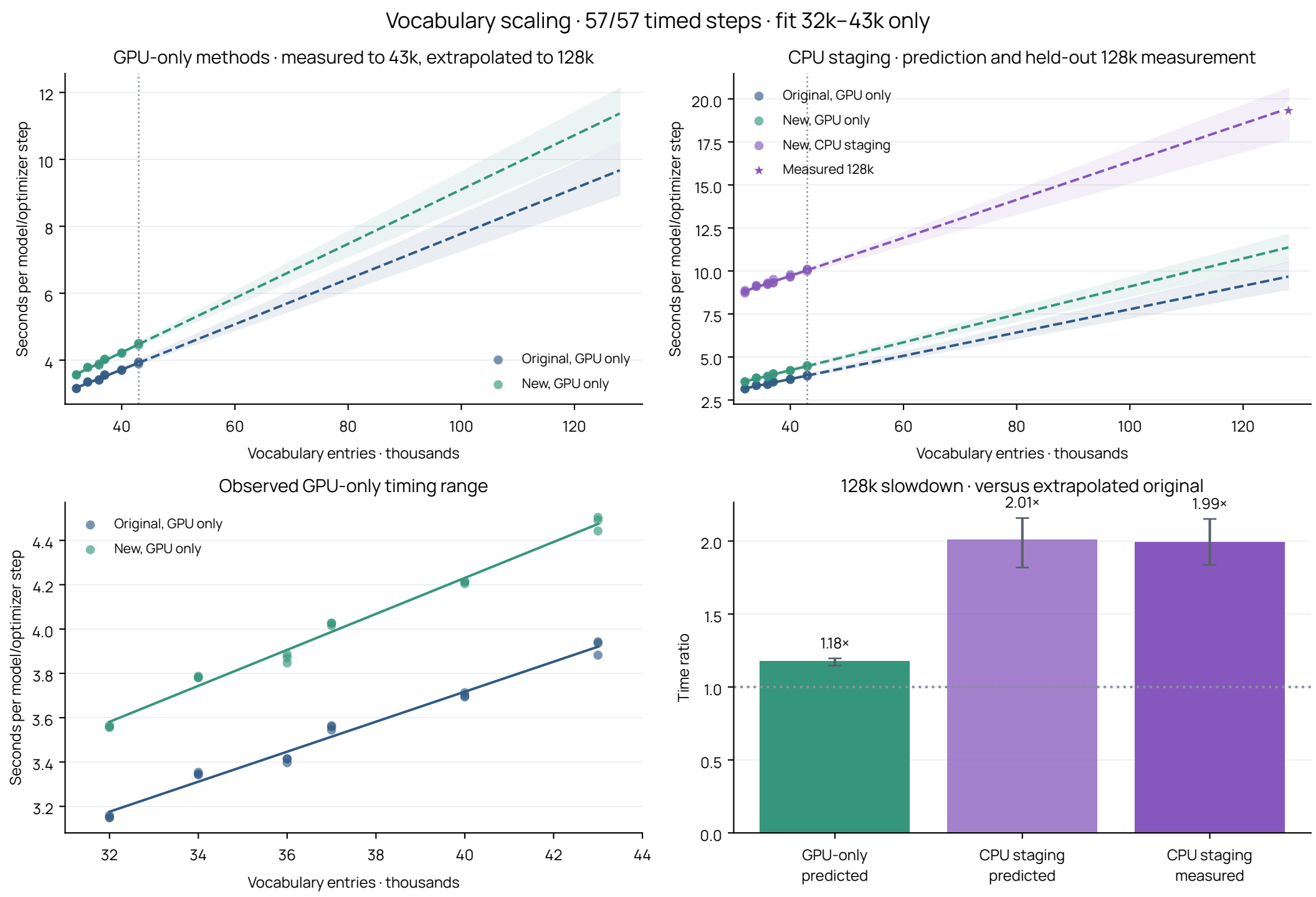
Vocabulary scaling · 57/57 timed steps · fit 32k–43k only
GPU-only methods · measured to 43k, extrapolated to 128k
Seconds per model/optimizer step
Vocabulary entries · thousands
Original, GPU only
New, GPU only
CPU staging · prediction and held-out 128k measurement
Original, GPU only
New, GPU only
New, CPU staging
Measured 128k
Seconds per model/optimizer step
Vocabulary entries · thousands
Observed GPU-only timing range
Original, GPU only
New, GPU only
Seconds per model/optimizer step
Vocabulary entries · thousands
128k slowdown · versus extrapolated original
Time ratio
1.18×
2.01×
1.99×
GPU-only predicted
CPU staging predicted
CPU staging measured